

# AQUACULTURE AND WATERBODY MONITORING SYSTEM

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**Abstract**—The field of aquaculture is amongst those that have experienced the greatest rate of growth within the industry of food production. The process of checking the water quality in such systems tends to be laborious and is associated with errors. Our project seeks to provide a means of monitoring water quality through an innovative system involving ASVs fitted with sensors for tracking key parameters like dissolved oxygen, redox potential, pH levels, temperature, turbidity, TDS, and conductivity. The ASVs monitor different positions within the pond using 3D technology and transmit data through IoT. Once the water quality starts degrading, alarm signals are sent to farmers in real-time allowing them to take appropriate measures to protect fish lives.

**Keywords**—*Waterbody management, App development, monitoring sensors, SDG Goals*

## I. INTRODUCTION

Aquaculture is an important sector when it comes to meeting the increasing demand of food worldwide, thus becoming one of the main sources of profit and protein for many individuals. But one of the key aspects in ensuring the proper functioning of these ecosystems is the continuous control over their water quality, which may be very hard to achieve because of the numerous negative consequences of bad water conditions such as disease, fish death, and low productivity. Unfortunately, conventional systems for water quality control have proven ineffective at achieving that goal.

To overcome these problems, we will suggest a water quality monitoring approach using Autonomous Surface Vehicles (ASVs) tailored for use in aquaculture. In contrast to immovable and stationary sensors, our ASV will be able to move autonomously through different water bodies like shrimp ponds and fish ponds, acquiring information about pH, turbidity, water temperature, and dissolved oxygen in real time.

The fixed sensors can measure parameters at certain locations only, thus restricting the comprehensive analysis of water conditions. On the other hand, in our system, the microcontroller works in association with the real-time sensors and sends the acquired information via wireless transmission to the cloud platform. The cloud-based information is accessible through web as well as mobile applications.

This innovation integrates the use of automated solutions, monitoring and analysis of data, and artificial intelligence to enable quick and informed decision-making by farmers. Whenever the parameters of the water are not within optimal ranges, an alert is generated immediately, allowing the farmer to take necessary actions to avoid any potential losses. This solution is easily customizable for varying water conditions.

In addition, this project will provide accurate information to aqua-farmers to manage their farms effectively and sustainably. In other words, there is less use of chemicals, less water wastage, and minimal environmental footprint. The innovative technology will help aqua-farmers to make sustainable decisions for their aquaculture farms in the long run.

In relation to SDGs 6, 12, and 14, our work contributes directly towards the sustainability of water and ecosystems. Through consistent monitoring of parameters such as pH, turbidity, and dissolved oxygen content, the project improves the quality of water and reduces any forms of pollution while maximizing utilization. It promotes sustainable aquaculture and consumption through waste minimization, prevention of overfishing, and efficient utilization of freshwater resources. The project also offers useful information for policy-makers, thus fostering stakeholder awareness and conservation of biodiversity.

## II. METHODOLOGY

The approach we take includes some important phases. These are, The first phase is the development of an interactive mobile application that acts as the primary medium through which the users will interact with the system. The application will have an efficient and simple layout.

The process of sensor integration entails mounting water quality sensors, such as those that measure pH, dissolved oxygen, and temperature, on a microcontroller. The sensors gather data and convert it into a transmittable format for the real-time monitoring of the aquaculture ponds' various conditions.

All data from the sensors will be transferred to the backend in the cloud using Firebase, ensuring that all information is synchronized in real time. This helps ensure that all data is secure, and the mobile application, cloud technology, and hardware can collaborate effectively and efficiently.

Data retrieved from Firebase is converted into comprehensible graphical representation. Graphs, along with other colored indicators, make it easier for the user to know the current status of the water. As a result, the user knows what needs to be done.

Live testing occurs in an actual aquaculture pond environment, where performance tests are conducted to assess the efficacy of the system under such environmental conditions. It helps verify whether the sensor readings are correct, data transmission is reliable, and the mobile application is responsive.

Post-field trial stage involves collecting feedback and recommendations from stakeholders and farmers. The collected feedback helps to make changes to the design of the interface, increase reliability of the system, and tailor the system to the practical requirements of its users.

## III. LITERATURE SURVEY

[1] According to the article, an intelligent water quality monitoring system that employs IoT technologies has been developed. Sensors monitor parameters such as pH, temperature, moisture in soil, and water flow, and the information is relayed in real-time through GSM and LoRa WAN. To ensure affordability and sustainability, the system is equipped with capacitive sensors and features a deep learning algorithm to verify whether the water quality is fit for human consumption.

[2] Here, the article talks about the application of IoT technology and cloud computing in the development of aquaculture. It mentions their importance in boosting productivity, saving money, and implementing smart solutions for water quality management and fish feeding. In this review, 17 kinds of sensors are mentioned, where temperature sensors, pH sensors, and dissolved oxygen sensors are the most common, while turbidity sensors, ammonia sensors, and conductivity sensors are the least common.

[3] A novel water quality monitoring system which utilizes a remote operated vehicle is introduced herein. This system monitors the quality of water by using sensors such as pH, dissolved oxygen, temperature and total dissolved solids sensors and transmits the information to different parties using various means of communications. For monitoring purposes, the system makes use of BetaTherm thermistors for measuring the temperature and ultrasonic sensors for the monitoring of water level/flow.

The IWMS-TAMARA system is another system that has been developed by the Technology Development Board under the AMRUT 2.0 Mission. The system is designed as a conservation tool for conserving water bodies using innovative IoT-based systems. The IWMS-TAMARA system facilitates circular water management using the Blue Economy theory. The system has an intelligent sedimentation aeration system that ensures there is enough oxygen in the water layers below the surface. There is a climate-based water quality monitoring system that manages nutrient and oxygen concentrations. The system also has a weed removal system called the PLASHBOT.

The Department of Fisheries Development and Fishermen's Welfare came into existence as a separate department in 1964, whereas, prior to that, it functioned as a part of the Department of Agriculture. It has been constituted with the primary objective of developing fisheries along with their conservation, and assisting fishermen and other people engaged in the field of fisheries through the utilization of all possible means..

Several scientific articles describe significant advancements made possible through the use of IoT technology, cloud computing, and sensor systems. The technologies such as IoT water monitoring system, IWMS-TAMARA, and the remote operated vehicle system have presented advanced solutions that are both effective and innovative in the areas of information gathering, feeding management, aeration, and pollution detection. These studies reveal the benefits brought by the use of the aforementioned technology in terms of efficiency, sustainability, accessibility, optimization of resources, and increased production in the fisheries. Sensing techniques measuring factors such as pH value, oxygen, temperature, turbidity, and conductance are crucial for environmental protection and aquaculture production.

These studies have yielded some common advantages which include: affordability, scalability, low power requirements, real-time monitoring, and enhanced efficiency in aquaculture. However, there are disadvantages associated with this technology as well. Most of these systems rely extensively on reliable communication networks; have high initial costs; need frequent calibration of sensors; and are not easily accessible to small farmers. Our project seeks to address these problems through a design of an inexpensive, portable, offline-based, modular and hybrid data gathering technique for fish farming.

#### IV. PROPOSED SOLUTION

This proposed idea may greatly revolutionize the aquaculture business through the employment of ASVs equipped with advanced sensors that are able to monitor water quality on a continuous basis. This novel idea will be made possible by the adoption of real-time data analysis technology along with artificial intelligence for sustainable agricultural practices.

The system architecture comprises several key components:

**Hardware:** In this system, the Autonomous Surface Vehicles (ASVs) have been equipped with highly sensitive sensors that monitor parameters of water quality such as dissolved oxygen levels, pH levels, temperature, turbidity, TDS (Total Dissolved Solids), and electric conductivity. The autonomous surface vehicles provide accurate and dependable data ensuring optimal conditions of water for aquaculture.

**Software:** The software facilitates data collection, processing, and analysis. By making use of artificial intelligence algorithms, the software provides information regarding growth rates and triggers warnings in case water quality begins to deteriorate. This ensures that the farmers can monitor their ponds effectively and act accordingly.

**Continuous Monitoring:** The ASVs monitor the water quality in ponds on an ongoing basis, issuing real-time warnings when any changes occur. With continuous monitoring, the farmers can identify any changes to the water conditions promptly and take action to ensure their aquatic organisms have a good living environment.

**Data Transmission:** With the help of IoT technology, the system transmits water quality data on a constant basis. With this continuous flow of data, it becomes easier for the farmers to receive immediate alerts if any change takes place in water quality.

**Visualization of Data:** The collected data will be shown on user-friendly dashboards which indicate the important metrics and growth trends. Through this process, it is expected that the farmers can easily observe the progress in terms of water quality as well as aquatic growth, which helps them to take sustainable decision..

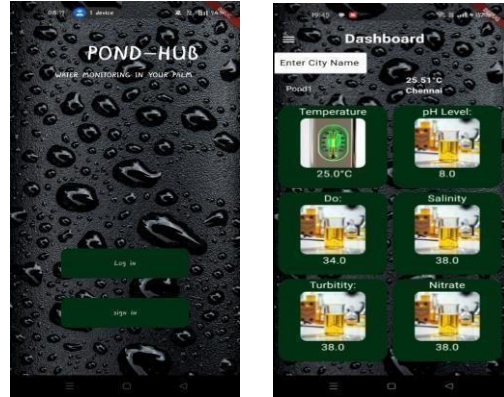


Fig. 2 Software Application

**How It Works:** There are sensors located at various points inside the pond that gather the information continuously and stream it to the Arduino board. Once the data reaches the Arduino board, it is sent through the internet to the ES6 module, which then uploads it to Firebase. In Firebase, each user has his/her own personal sheet containing the data, which gets updated in real-time. The app's API retrieves the information from Firebase and presents it on the home page, while Syncfusion generates charts to present the data. The users can also export the data for a particular time period into Excel files.

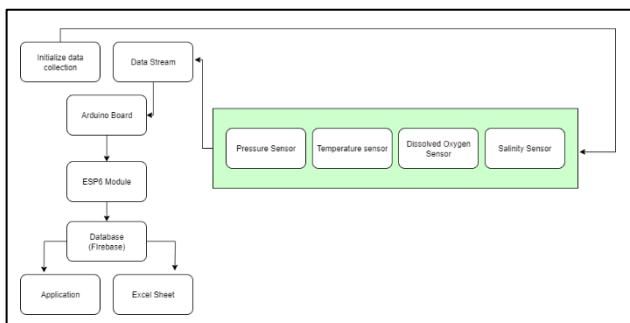


Fig. 1. Architectural Diagram

## V. NAVIGATION ALGORITHM

The A\* or A-Star algorithm is an efficient method for the Autonomous Surface Vehicle (ASV) to navigate in the aquaculture pond. The algorithm calculates the optimal path that is also the shortest path from one point to another by evaluating the distance covered so far and the distance yet to be covered.

**The algorithm makes use of the formula below to determine the optimal route:**

$$f(n) = g(n) + h(n)$$

**What does this imply?**

- **f(n)** represents the cost of getting to a certain node.
- **g(n)** is the cost of getting to the current node from the initial node.
- **h(n)** is the cost estimate of getting to the destination node from the current node.

**The two values are added to determine the optimal route.**

The heuristics (h) used for estimating the distance to the goal include:

- Manhattan distance (if there are 4 possible directions on a grid)

$$h(n) = |x_1 - x_2| + |y_1 - y_2|$$

- Euclidean distance (distance between two points as the crow flies)

$$h(n) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

- Zero-heuristic function ( $h=0$ ). In that case, A\* search behaves

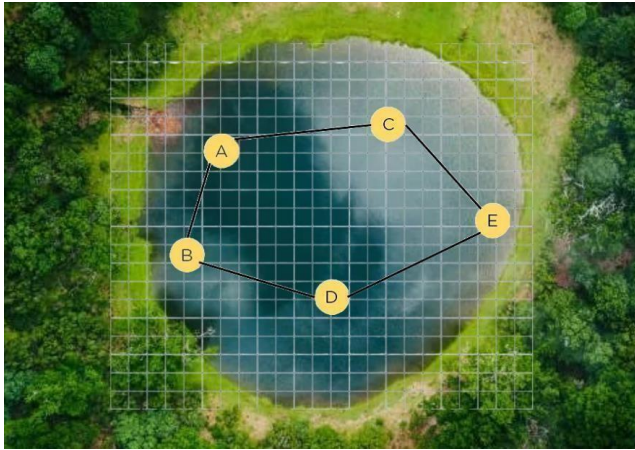


Fig.3 working of A\* Algorithm

- Make two sets:
- Open list – nodes to be processed (initially containing the initial node).
- Closed list – nodes that have been processed.
- Add the initial node to the open list, assigning its total cost as  $f(\text{initial node}) = h(\text{initial node})$ .
- Until the open list is not empty:
- Choose the node n with minimum total cost  $f(n)$ .
- If n is the goal, then the path is discovered – start reconstructing the path.
- Otherwise, add n to the closed list.
- For each neighbor of n:

- Calculate  $g(\text{neighbour}) = g(n) + \text{cost}(n, \text{neighbour})$ .
- Calculate  $h(\text{neighbour})$  (heuristic, e.g., Euclidean or Manhattan distance).
- Set  $f(\text{neighbor}) = g(\text{neighbor}) + h(\text{neighbor})$ .
- Add neighbor to open set if it is not in open set or its f value is lower.

2. Repeat until goal is reached or open set is empty.

The A\* algorithm assists the Autonomous Surface Vehicle (ASV) to navigate in a more effective and precise manner. This is done by balancing the cost function (g), which is the distance that has already been covered, and heuristic function (h), which estimates the distance still left to reach the target location. Thus, the ASV becomes capable of finding the optimal path without any deviation due to the presence of obstacles such as buoys, aquatic weeds, or other boats..

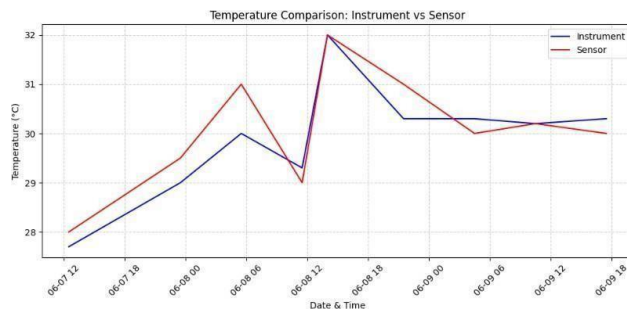
## VI. TESTING

Static product modeling along with its performance analysis was conducted under laboratory conditions in order to assess its reliability and effectiveness. This was done under artificial shrimp farming conditions at CIBA, ECR, Chennai. As all critical factors were already determined to industrial standards, this made the process of testing and calculation easier.

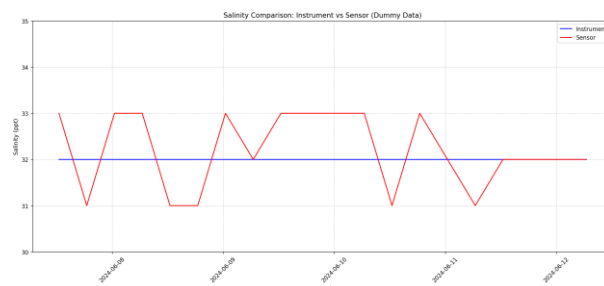
| S.No | Date         | Time     | Instrument |      |      |          | Sensor |     |      |          |
|------|--------------|----------|------------|------|------|----------|--------|-----|------|----------|
|      |              |          | DO         | pH   | TEMP | SALINITY | DO     | pH  | TEMP | SALINITY |
| 1    | 07 June 2024 | 12:30 PM | 6.62       | 7.62 | 27.7 | 32       | 6.75   | 8.2 | 28   | 32       |
| 2    |              | 11:30 PM | 4.4        | 7.8  | 30   | 32       | 4.17   | 7.5 | 29   | 33       |
| 3    | 08 June 2024 | 5:30 AM  | 5.3        | 7.4  | 29.3 | 32       | 5.51   | 7.4 | 29.7 | 33       |
| 4    |              | 11:30 AM | 6.9        | 7.4  | 32   | 32       | 6.26   | 7.4 | 32.9 | 32       |
| 5    |              | 7:00 PM  | 6.21       | 7.4  | 31.7 | 32       | 6.54   | 7.3 | 32   | 32       |
| 6    | 09 June 2024 | 9:30 PM  | 4.44       | 7.4  | 30.5 | 32       | 4.68   | 7.3 | 31.2 | 32       |
| 7    |              | 5:30 AM  | 4.86       | 7.4  | 29.4 | 32       | 4.91   | 7.4 | 30.3 | 32       |
| 8    |              | 10:30 AM | 7.91       | 7.4  | 32   | 32       | 7.5    | 7.4 | 31.2 | 32       |
| 9    | 12 June 2024 | 9:58 AM  | 5.06       | 7.7  | 30.2 | 32       | 5.3    | 7.5 | 30.7 | 33       |
| 10   |              | 11:32 AM | 6.03       | 7.7  | 30.8 | 32       | 6.07   | 7.4 | 31.5 | 33       |
| 11   |              | 6:23 PM  | 4.50       | 7.6  | 31.2 | 32       | 4.29   | 7.4 | 32.2 | 32       |
| 12   | 13 June 2024 | 9:08 PM  | 4.28       | 7.6  | 31   | 32       | 4.12   | 7.4 | 31.5 | 31       |
| 13   |              | 9:05 AM  | 5.63       | 7.6  | 30   | 32       | 5.55   | 7.3 | 29.7 | 31       |
| 14   |              | 7:18 PM  | 3.77       | 7.7  | 30   | 32       | 3.8    | 7.4 | 29.7 | 31       |
| 15   | 14 June 2024 | 9:55 PM  | 3.10       | 7.7  | 29.6 | 32       | 3.6    | 7.5 | 29.4 | 33       |
| 16   |              | 9:18 AM  | 4.88       | 7.7  | 29   | 32       | 4.5    | 7.5 | 29.7 | 33       |
| 17   |              | 11:15 AM | 6.73       | 7.6  | 31   | 32       | 7.1    | 7.4 | 30   | 33       |
| 18   | 15 June 2024 | 6:17 PM  | 5.36       | 7.6  | 30.9 | 34       | 4.96   | 7.4 | 31.5 | 33       |
| 19   |              | 11:55 PM | 3.8        | 7.6  | 30.5 | 34       | 3.5    | 7.3 | 29.1 | 32       |
| 20   |              | 11:08 AM | 5.00       | 7.6  | 29.6 | 32       | 4.8    | 7.5 | 31.5 | 33       |
| 21   | 16 June 2024 | 1:45 PM  | 6.3        | 7.8  | 30.5 | 32       | 6.6    | 7.6 | 29.7 | 32       |
| 22   |              | 6:02 PM  | 5.43       | 7.8  | 30.4 | 32       | 5.6    | 7.5 | 30.9 | 32       |
| 23   |              | 9:27 PM  | 3.54       | 7.8  | 29.7 | 32       | 3.9    | 7.5 | 31   | 32       |
| 24   | 17 June 2024 | 10:30 AM | 7.68       | 7.8  | 29.9 | 32       | 7.1    | 7.5 | 32.5 | 32       |
| 25   |              | 12:46 AM | 8.38       | 7.8  | 31.2 | 32       | 8      | 7.6 | 33   | 31       |
| 26   |              | 7:10 PM  | 4.6        | 7.8  | 30.1 | 32       | 4.2    | 7.6 | 31.3 | 32       |
| 27   | 18 June 2024 | 9:21 PM  | 3.98       | 7.8  | 31.3 | 32       | 3.59   | 7.7 | 32.1 | 31       |
| 28   |              | 11:00 AM | 8.44       | 7.8  | 30.5 | 32       | 8.1    | 7.6 | 32.2 | 32       |
| 29   |              | 12:45 PM | 9.43       | 7.8  | 31.2 | 32       | 9.5    | 7.6 | 32.5 | 32       |
| 30   | 19 June 2024 | 7:36 PM  | 4.6        | 7.7  | 30.8 | 32       | 4.34   | 7.6 | 31.9 | 32       |
| 31   |              | 9:35 PM  | 3.8        | 7.7  | 31   | 32       | 3.7    | 7.6 | 31.2 | 32       |
| 32   |              | 10:57 AM | 7.88       | 7.7  | 29.6 | 32       | 7.55   | 7.5 | 28.2 | 32       |
| 33   | 19 June 2024 | 1:33 PM  | 8.65       | 7.7  | 30.4 | 32       | 8.44   | 7.6 | 29.1 | 31       |
| 34   |              | 6:00 PM  | 6.16       | 7.8  | 31   | 32       | 5.88   | 7.6 | 29.4 | 32       |
| 35   |              | 9:09 PM  | 3.68       | 7.8  | 30.5 | 32       | 3.8    | 7.6 | 29.8 | 31       |
| 36   | 19 June 2024 | 10:52 AM | 8.30       | 7.8  | 29.6 | 32       | 7.97   | 7.7 | 27.9 | 31       |
| 37   |              | 1:00 PM  | 8.89       | 7.9  | 30.2 | 32       | 8.4    | 7.8 | 28.5 | 32       |
| 38   |              | 5:14 PM  | 6.37       | 7.9  | 29.6 | 32       | 5.9    | 7.8 | 28.8 | 32       |
| 39   |              | 7:09 PM  | 3.99       | 7.9  | 30.1 | 32       | 3.75   | 7.7 | 29.1 | 32       |

|    |              |          |      |     |      |    |      |     |       |    |
|----|--------------|----------|------|-----|------|----|------|-----|-------|----|
| 40 |              | 10:19 AM | 7.44 | 7.8 | 29   | 32 | 7.17 | 7.6 | 27.6  | 31 |
| 41 | 20 June 2024 | 2:30 PM  | 9.54 | 7.8 | 30.9 | 32 | 9.15 | 7.7 | 30    | 32 |
| 42 |              | 7:30 PM  | 3.54 | 7.8 | 30.6 | 32 | 3.72 | 7.7 | 29.7  | 31 |
| 43 |              | 9:20 PM  | 3.88 | 7.8 | 30.9 | 32 | 4.01 | 7.7 | 29.1  | 31 |
| 44 |              | 7:50 AM  | 4.37 | 7.7 | 29   | 32 | 4.1  | 7.6 | 27.9  | 32 |
| 45 | 21 June 2024 | 11:57 AM | 6.32 | 7.7 | 30.3 | 32 | 6.08 | 7.5 | 28.8  | 32 |
| 46 |              | 5:08 PM  | 5.18 | 7.8 | 31.5 | 32 | 5.32 | 7.6 | 30    | 31 |
| 47 |              | 7:33 PM  | 3.78 | 7.8 | 30.9 | 32 | 3.98 | 7.6 | 29.5  | 31 |
| 48 |              | 2:42 PM  | 7.2  | 7.7 | 31.8 | 32 | 7.53 | 7.5 | 32.1  | 32 |
| 49 | 22 June 2024 | 4:04 PM  | 5.1  | 7.7 | 31.1 | 32 | 4.84 | 7.6 | 31.5  | 31 |
| 50 |              | 7:01 PM  | 4.21 | 7.7 | 30.2 | 32 | 4.5  | 7.5 | 30.8  | 31 |
| 51 |              | 9:52 PM  | 4.4  | 7.6 | 30.5 | 32 | 4.13 | 7.7 | 29.89 | 31 |
| 52 |              | 7:00 AM  | 4.51 | 7.8 | 29.6 | 32 | 4.38 | 7.6 | 30.05 | 32 |
| 53 | 23 June 2024 | 5:45 PM  | 5.9  | 7.7 | 29.5 | 32 | 6.3  | 7.7 | 30.98 | 32 |
| 54 |              | 7:07 PM  | 3.84 | 7.7 | 31.4 | 32 | 4.2  | 7.6 | 29.86 | 31 |
| 55 |              | 9:45 PM  | 4.01 | 7.7 | 30.8 | 32 | 3.88 | 7.5 | 30.76 | 33 |
| 56 |              | 12:24 PM | 5.58 | 7.8 | 30   | 32 | 5.31 | 7.6 | 30.77 | 31 |
| 57 | 24 June 2024 | 2:26 PM  | 7.3  | 7.8 | 30.8 | 32 | 6.93 | 7.5 | 29.8  | 32 |
| 58 |              | 4:26 PM  | 5.85 | 7.9 | 31.1 | 32 | 6.1  | 7.7 | 30.9  | 31 |
| 59 |              | 8:54 PM  | 4.2  | 7.9 | 30.1 | 32 | 4.42 | 7.6 | 29.6  | 33 |
| 60 |              | 12:24 PM | 6.12 | 7.8 | 30.9 | 32 | 6.45 | 7.7 | 29.78 | 32 |
| 61 | 25 June 2024 | 2:45 PM  | 7.12 | 7.7 | 31.5 | 32 | 6.82 | 7.7 | 31.18 | 31 |
| 62 |              | 7:20 PM  | 3.98 | 7.8 | 29.8 | 32 | 4.37 | 7.6 | 31.16 | 32 |
| 63 |              | 10:40 PM | 4.2  | 7.8 | 30.5 | 32 | 4.08 | 7.7 | 31.04 | 31 |
| 64 |              | 1:09 PM  | 6.68 | 7.8 | 30.9 | 32 | 6.38 | 7.9 | 29.65 | 31 |
| 65 | 26 June 2024 | 3:10 PM  | 7.28 | 7.9 | 31.5 | 32 | 6.83 | 7.8 | 31.05 | 32 |
| 66 |              | 5:19 PM  | 5.7  | 7.9 | 30.8 | 32 | 5.52 | 7.6 | 30.09 | 32 |
| 67 |              | 3:25 PM  | 6.52 | 7.8 | 29.8 | 32 | 6.8  | 7.5 | 30.8  | 33 |
| 68 |              | 6:15 PM  | 5.50 | 7.7 | 31.1 | 32 | 5.33 | 7.9 | 30.5  | 31 |
| 69 | 28 June 2024 | 8:15 PM  | 4.4  | 7.7 | 29.5 | 32 | 4.2  | 7.6 | 30.3  | 32 |
| 70 |              | 10:15 PM | 3.85 | 7.9 | 29.4 | 32 | 4.1  | 7.6 | 31.2  | 31 |
| 71 |              | 11:57 AM | 5.9  | 7.7 | 30.8 | 32 | 5.65 | 7.9 | 29.6  | 32 |
| 72 |              | 2:00 PM  | 6.3  | 7.7 | 31.4 | 32 | 6.55 | 7.6 | 30.8  | 31 |
| 73 | 29 June 2024 | 3:45 PM  | 6.62 | 7.9 | 30.5 | 32 | 6.4  | 7.7 | 29.4  | 31 |
| 74 |              | 10:00 AM | 4.89 | 7.9 | 31.2 | 32 | 5.03 | 7.7 | 30.63 | 31 |
| 75 |              | 1:10 PM  | 5.9  | 7.9 | 30.9 | 32 | 6.1  | 7.6 | 29.94 | 32 |
| 76 |              | 3:10 PM  | 6.3  | 8.0 | 31.1 | 32 | 6.0  | 7.8 | 30.2  | 32 |
| 77 |              | 5:20 PM  | 5.8  | 8.0 | 30.5 | 32 | 5.61 | 7.7 | 29.6  | 31 |

Table1.Testing data

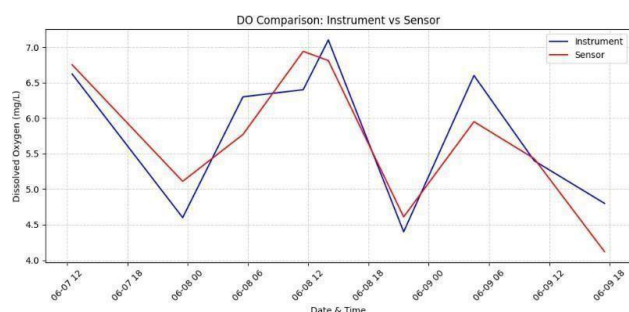


Graph3.Temperature comparison chart

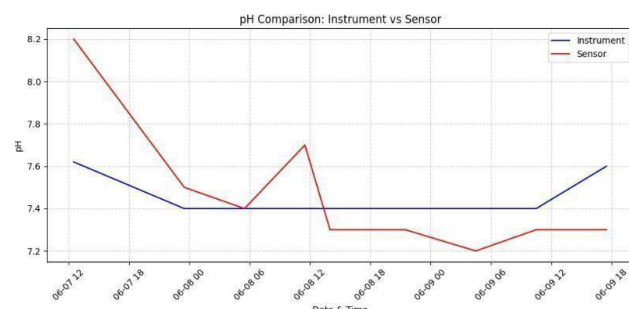


Graph4.Salinity comparison chart

## VII. COMPARATIVE ANALYSIS



Graph1.Dissolved oxygen comparison chart



Graph2.pH comparison chart

The comparison of the sensor data and that from professional equipment revealed a fairly good correspondence for all the four parameters tested. With respect to the first parameter – the dissolved oxygen (DO) level – on average, the sensor differed by 6-8% from the instrument reading. This figure could be considered quite an accurate one although some discrepancies existed when it came to low DO levels. In turn, the pH values measured by means of the sensor differed by no more than 3-5%, showing thus that the sensor is highly reliable. What is more, temperature readings demonstrated the highest degree of similarity and accuracy – the variations did not exceed 2%. At the same time, salinity figures were also relatively close and varied from 1-3%, whereas the instrument readings did not change at all.

## VIII. CONCLUSION

Overall, this technology is innovative and very helpful in sustainable aquaculture since it allows the use of Autonomous Surface Vehicles to continuously monitor water quality. The system uses efficient sensors, real-time data transmission, analysis with the help of artificial intelligence, and visualization of data which makes it more useful for farmers. The use of technologies such as the A\* algorithm for navigation and the Internet of Things (IoT) for communication allows increasing precision and efficiency in the management of ponds. Ultimately, this method helps ensure better growth and health conditions for aquatic organisms while decreasing errors and labor efforts of workers.

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